ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NSMCE2-DEFICIENT BREAST CANCER CELLS

A Dissertation submitted by

Ms. Manohari M Kulkarni

(US No: NU24BSCB03)

To



(Estd. under Section 3, UGC Act 1956)
(Placed under Category 'A' by MHRD., Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of

MASTER OF SCIENCE (CANCER BIOLOGY)

DEPARTMENT OF MOLECULAR GENETICS AND CANCER

NITTE UNIVERSITY CENTRE FOR SCIENCE EDUCATION & RESEARCH

DERALAKATTE, MANGALURU-575018

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CERTIFICATE

This is to certify that the dissertation titled "ROLE OF COMPENSATORY SUMOLYATION MECHNAISMS IN NSMCE2-DEFICIENT BREAST CANCER CELLS" is a bonafide research work carried out by **Manohari M Kulkarni, NU24BSCB03** under the guidance of **Dr. Vivek Tripathi** and co-guidance of **Dr. Sudarshan Kini** in the Department of Molecular genetics and Cancer of **Nitte University Centre of Science Education and Research**

The same is being submitted to the Nitte (Deemed to be University) in partial fulfillment of the requirements for the award of Master of Science **Cancer Biology**.

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DECLARATION

I, **Manohari M Kulkarni (NU24BSCB03),** hereby declare that the Dissertation work titled "**ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NSMCE2-DEFICIENT BREAST CANCER CELLS**" is my original work and has been carried out under the guidance of Dr. **Vivek Tripathi, Department of Molecular Genetics and Cancer, Nitte University Centre for Science Education and Research** and Co-guidance of **Dr. Sudarshan Kini, Department of Bio and Nano Technology, Nitte University Centre for Science Education and Research** is being submitted to the Nitte (Deemed to be University) in partial fulfilment of the requirements for the award of Master of **Cancer Biology**

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Mangalore **Ms. Manohari Kulkarni**

May, 2026

LIST OF ABBREVIATIONS

Sr. no	Abbreviation	Extended form
1.	°C	Degree Celsius
2.	µg	Micrograms
3.	µL	Micro liter
4.	BSA	Bovine Serum albumin
5.	cDNA	Complementary DNA
6.	CO_2	Carbon dioxide
7.	DMEM	Dulbecco's Modified Eagle Medium
8.	DSB	Double-strand DNA breaks
9.	ER	Estrogen receptor
10.	FBS	Fetal Bovine Serum
11.	HER2	Human Epidermal Growth Factor Receptor 2
12.	LN_2	Liquid nitrogen
13.	mg	Milligrams
14.	mL	Milliliter
15.	MMS21	Methyl methanesulfonate 21

16.	NSE	Non-SMC element
17.	NSMCE2	Non-SMC element 2
18.	PARP	Poly ADP ribose phosphate
19.	PARPi	Poly ADP ribose phosphate inhibitor
20.	PBS	Phosphate buffer saline
21.	PBST	Phosphate Buffered Saline with Tween-20
22.	PCR	Polymerase chain reaction
23.	PIC	Protease Inhibitor Cocktail
24.	PMSF	Phenylmethanesulfonyl fluoride
25.	PR	Progesterone receptor
26.	RT-qPCR	Real-time quantitative Polymerase chain reaction
27.	SDS-PAGE	Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis

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ABSTRACT

Despite many advancements in the treatment of cancers, breast cancer continues to be a major source of death due to cancer around the globe. Triple Negative Breast Cancer (TNBC), which is known for being aggressive in nature and having early relapse, is the most difficult subtype of breast cancer for treatment because of its resistance to current treatments. While there is no hormone and HER2 receptor on this subtype of breast cancer, chemotherapy and PARP inhibitors remain some of the few treatment options that can work against it. What makes treatment ineffective in such cases is TNBC cells' ability to alter their DNA damage response (DDR) system, which is known to depend on post-translational modifications like SUMOylation that manages the recruitment of repair proteins in damaged DNA. The protein involved in SUMOylation in the SMC5/6 chromosomal maintenance complex is NSMCE2, whose role in therapy resistance has not been explored in this type of breast cancer.

The aim of the current study was to close this gap by performing expression profiling of NSMCE2 in two TNBC cell lines (MDA-MB-231 and MDA-MB-468) compared with a control line expressing estrogen receptors (MCF-7) and by investigating the effects of the Olaparib treatment on the cellular migration and viability of the TNBC lines. The results showed that the level of the NSMCE2 gene and protein expression in both cell lines was higher than the positive control (MCF-7). Karyotypic analysis in all cell lines indicated an abnormal number of hyper triploid chromosomes per metaphase spreads. Migration of cells in all three lines was suppressed by the Olaparib treatment, but this effect was not as strong in the case of the MDA-MB-231 line.

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CHAPTER I

INTRODUCTION

1.1 PREAMBLE

Breast cancer has become the most prevalent malignancy in the world, overtaking lung cancer, with about 2.3 million cases every year. Of particular concern is Triple Negative Breast Cancer (TNBC) that is distinguished by the lack of the presence of ER (estrogen receptor), PR (progesterone receptor), and HER2(1). It makes up 15-20% of all breast cancers with the characteristics of an aggressive behavior pattern, a high risk of relapse in the initial three to five years after diagnosis, and the chances of visceral metastases that occur mainly in the brain and the lungs. TNBC demonstrates high genomic instability, due to the DDR deficiencies and "BRCAness" phenomenon, leading to an impairment of the precise repair of DNA by homologous recombination. This instability is responsible for both tumor heterogeneity and the evolution, but it also means vulnerability to the use of platinum drugs and PARPi (PARP inhibitors). Tumor cells often adapt to this through metabolic reprogramming and adaptation to their DDR system by stabilizing replication forks with other mechanisms, which causes treatment resistance and relapse of the disease(2).

However, an understudied regulator involved in this process is the Structural Maintenance of Chromosomes 5/6 (SMC5/6) complex, which plays an essential role in preserving genomic stability during replication stress (3). NSMCE2 is one component of the SMC5/6 complex, serving as the SUMO E3 ligase (4). SUMOylation mediated by NSMCE2 offers a connection between chromosomal structural maintenance and DNA repair proteins dynamics. Although complete absence of NSMCE2 results in cancer susceptibility, new evidence shows that part of its tumor suppressor function may be independent of its ligase activity, suggesting multifunctional properties for the complex (5).

Post-translational modifications (PTMs), particularly SUMOylation, regulate the addition and stabilization of repair factors at stalled forks, allowing DNA replication restart to occur efficiently. For NSMCE2-deficient TNBC cells experiencing higher genotoxic damage that would be lethal, it is proposed that there may be alternative SUMOylation pathways employed for their survival. Multiple studies have reported extensive alterations of the SUMOylation pathway in the presence of replication stress. When NSMCE2 function is compromised, cancer cells may upregulate alternative E3 ligases such as PIAS1 or PIAS4 to maintain the SUMOylation of essential repair proteins like BRCA1, 53BP1, or CtIP (3,4). This adaptive response potentially stabilizes replication forks and preserves DNA repair capacity, thereby driving chemoresistance against the very agents designed to exploit DDR vulnerabilities.

Understanding this, helps us identify how TNBC cells rewire their SUMOylation networks and reveal novel therapeutic targets to sensitize aggressive tumors to existing chemotherapies.

1.2 AIM AND OBJECTIVES

Aim: To investigate the role of compensatory SUMOylation mechanisms in maintaining DNA repair capacity and promoting chemoresistance in NSMCE2-deficient triple-negative breast cancer (TNBC) cells.

Objectives:

1. To downregulate NSMCE2 expression in MDA-MB-231 cells using siRNA transfection and confirm knockdown efficiency.
2. To investigate the effects of NSMCE2 downregulation on key cellular processes

1.3 LITERATURE REVIEW

1.3.1 BREAST CANCER AND CURRENT TREATMENT STRATEGIES

Breast cancer is a diverse disease in terms of histologic types, natural history, clinical behavior, and response to treatment. It is the most frequently diagnosed malignancy in women and remains a biologically heterogenous disease with clinically and molecular distinct subtypes with different disease progression patterns, therapeutic vulnerabilities, and patterns of relapse ((5,6). Among these, triple-negative breast cancer (TNBC) is defined by the absence of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2), which excludes endocrine therapy and HER2-directed therapy and leaves cytotoxic chemotherapy, immunotherapy and a limited number of biomarker-guided targeted agents as routine treatment options. TNBC is characterized by high histological grade, early recurrence, metastasis, and poor clinical outcome. (5–7)

Currently, the management of TNBC relies on a multimodal approach combining surgery, radiotherapy, and systemic therapy tailored to disease stage and tumour biology. Surgical resection, either breast-conserving surgery or mastectomy with axillary evaluation, forms the foundation of early-stage treatment, with the degree of residual disease following neoadjuvant therapy serving as a key prognostic indicator, while adjuvant radiotherapy further combines locoregional control given TNBC's well-documented propensity for local relapse (8,9). Anthracycline and taxane-based regimens such as AC-T remain the cytotoxic backbone of neoadjuvant and adjuvant treatment, achieving pathological complete response (pCR) rates of 30–40%, and platinum salts including cisplatin and carboplatin are increasingly incorporated for homologous recombination-deficient tumours, with the GeparSixto and BrighTNess trials demonstrating significantly improved pCR rates with carboplatin addition (10–12). Given TNBC's immunogenic character, immune checkpoint blockade has

become an important component of treatment, with the KEYNOTE-522 trial establishing pembrolizumab plus chemotherapy as the standard neoadjuvant regimen for high-risk early TNBC, and the IMpassion130 trial supporting atezolizumab plus nab-paclitaxel in PD-L1-positive metastatic disease (13,14) For germline BRCA1/2 mutation carriers, PARP inhibitors olaparib and talazoparib exploit synthetic lethality to induce tumour cell death, with the OlympiAD, EMBRACA, and OlympiA trials establishing their roles in both metastatic and adjuvant settings respectively (15–17). The antibody-drug conjugate sacituzumab govitecan, for heavily pretreated metastatic disease (18). These treatment strategies have improved outcomes for selected patient subsets; however, progression control remains limited because TNBC rapidly develops intrinsic or acquired resistance through clonal heterogeneity, adaptive stress signaling, altered drug efflux, immune evasion, restoration of DNA repair competence and replication fork stabilization (2,19,20). This provides a central rationale for investigating DNA repair plasticity and post-translational signaling that enables tumor cells to survive chemotherapy.

1.3.2 DNA REPLICATION STRESS

One of the important factors shaping TNBC behaviour is replication stress. Replication stress refers to the slowing or stalling of DNA synthesis caused by oncogene activation, nucleotide depletion, DNA lesions, transcription-replication conflicts, aberrant origin firing, difficult-to-replicate chromatin structures, or defects in DNA repair pathways (19,21,22). TNBC is especially vulnerable to this type of stress because of its higher proliferation rate, frequent *TP53* loss, defective G1/S checkpoint control, and chromosomal instabilities place a constant pressure on S-phase that leads to replication fork stall (5,6,19). When replication forks stalls, cells try to preserve genome integrity through ATR-CHK1 signaling, fork reversal, repriming, homologous recombination and protection of nascent DNA from nucleolytic

degradation (19,21,23). If these responses fail, stalled forks collapse into DNA double-strand breaks, promoting copy number changes, chromosome fragmentation and structural rearrangements.

“Fork-stabilized resistance” represents itself as a significant problem in terms of different genetic background of TNBC cell lines MDA-MB-231 and MDA-MB-468. Though both of the cell lines represent the characteristics of TNBC cells, the mechanism of coping with replication stress in the absence of BRCA1/2 deficiency is unique for both of them. Replication stress in TNBC cell lines MDA-MB-231 and MDA-MB-468 relies on their mutational contexts, and correlates with loss of G1/S checkpoint control.

- This stress is further exacerbated in MDA-MB-468 cells by the amplification of the EGFR gene and the absence of RB1 that rushes cells into S-phase and raises the probability of transcription-replication conflicts. (24)
- Alternatively, MDA-MB-231 cells live in replication stress through an oncogenic KRAS signaling which changes the timing of replication origin firing and initiates fork protection to prevent accumulation of structural rearrangements. (25)

Therefore, the behavior of DNA damaged breast cancer cells is not directed by a single pathway, but rather by the entire system of pathways or “toolbox” involved in the maintenance of the genome of the breast cancer cell.

1.3.3 SMC5/6 AND NSMCE2 IN CHROMOSOME MAINTAINANCE

The SMC5/6 complex has developed into a major player in the regulation of chromosome dynamics, replication-associated repair and genome stability in this network of genome maintenance. The structural maintenance of the chromosomes (SMC) family consists of cohesin, condensin and the SMC5/6 complex in each family

which is responsible for structuring chromatin and employing various functions within the chromosomes (26-28). The SMC5/6 complex plays a special role in DNA replication and repair mechanisms, such as replication fork recovery, resolution of recombination intermediates and maintenance of chromosomal integrity when a replication fork becomes stalled during stress. Also non-SMC element 2 (NSMCE2) which is another non-SMC subunit of the SMC5/6 complex and contains a SP-RING domain with SUMO E3 ligase activity (29, 30). Earlier we have shown that the human ortholog MMS21/NSMCE2 is a requirement for efficient repair of the genome and that it SUMOylates factors associated with genome surveillance. NSMCE2 is a key molecule connecting the architecture of chromosomes and the postnatal modifications of signalling

NSMCE2 is more than an accessory to repair. NSMCE2 also is involved in repair of lesions occurring during replication, handle recombination intermediate and maintain the integrity of the chromosome during mitosis (24,25). The impaired chromosome segregation, DNA structure overload and genotoxic stress sensitivity that results from the disruption of NSMCE2 or the SMC5/6 complex are all features of genomic instability. Few studies also propose roles of NSMCE2 in tumour suppression and tissue homeostasis, which can be separable from its SUMO ligase function proposing a variety of structural and enzymatic functions leading to biological output (24). NSMCE2-deficiency leads to impaired synthesis of structural genome, which leaves tumor cells vulnerable to alternative post-translational modifications or alternate parallel DNA damage response systems to maintain their viability in the presence of replication stress.

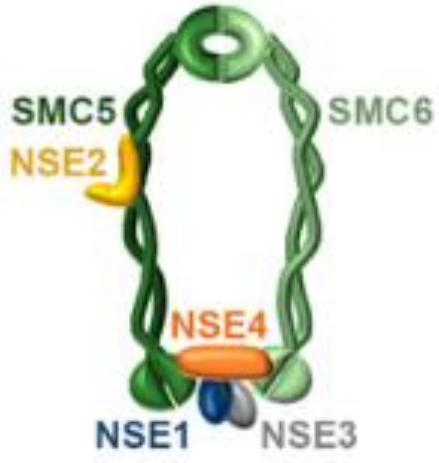


Fig 1.1: THE SMC5/6 complex and its components (26)

1.3.4 THE SUMOylation pathway

Small ubiquitin-like modifier (SUMO) proteins undergo covalent binding to lysine residues of substrate proteins in an ATP-dependent manner, via an enzyme cascade; this covalent modification is termed SUMOylation(23,27–29). Mammalian cells typically use SUMO1 in its monomeric form as a modifier, while SUMO2 and SUMO3 can form polymeric chains that can provide binding surfaces for SUMO-interacting motifs and for SUMO-targeted ubiquitin ligases (27,28)

The SUMOylation cycle starts with the activating enzyme heterodimer SAE1/SAE2 that adenylates mature SUMO to generate the thioester intermediate. However, the SUMO is activated and then transferred to a single E2 conjugating enzyme, UBC9, that recognises consensus motifs in substrates, and catalyses SUMO conjugation directly or via the action of SUMO ligases such as NSMCE2, PIAS-family and RanBP2 etc. The pathway is dynamic due to the degrading effect of SUMO-specific proteases (SENPs) which rapidly remove SUMO from substrates, which often involves transient, context-dependent remodeling of protein complexes (23,29,30)

In terms of function, SUMOylation is very rarely a binary mechanism of on/off. It rather, alters protein-protein interactions, position of protein in the cell, chromatin binding,

enzyme activity and assembling of macromolecules (23,29,30). A concept called “protein group SUMOylation” is a mechanism in which more than one neighbouring protein is simultaneously SUMO-modified to build a cooperative microenvironment for the stabilization of chromatin associated processes and for DNA repair (27), In the case of replication stress in which the coordination and exchange of many factors at damaged DNA forks efficiently play a key role in deciding whether cells undergo repair, checkpoint arrest, collapse and apoptosis, this model is applicable.

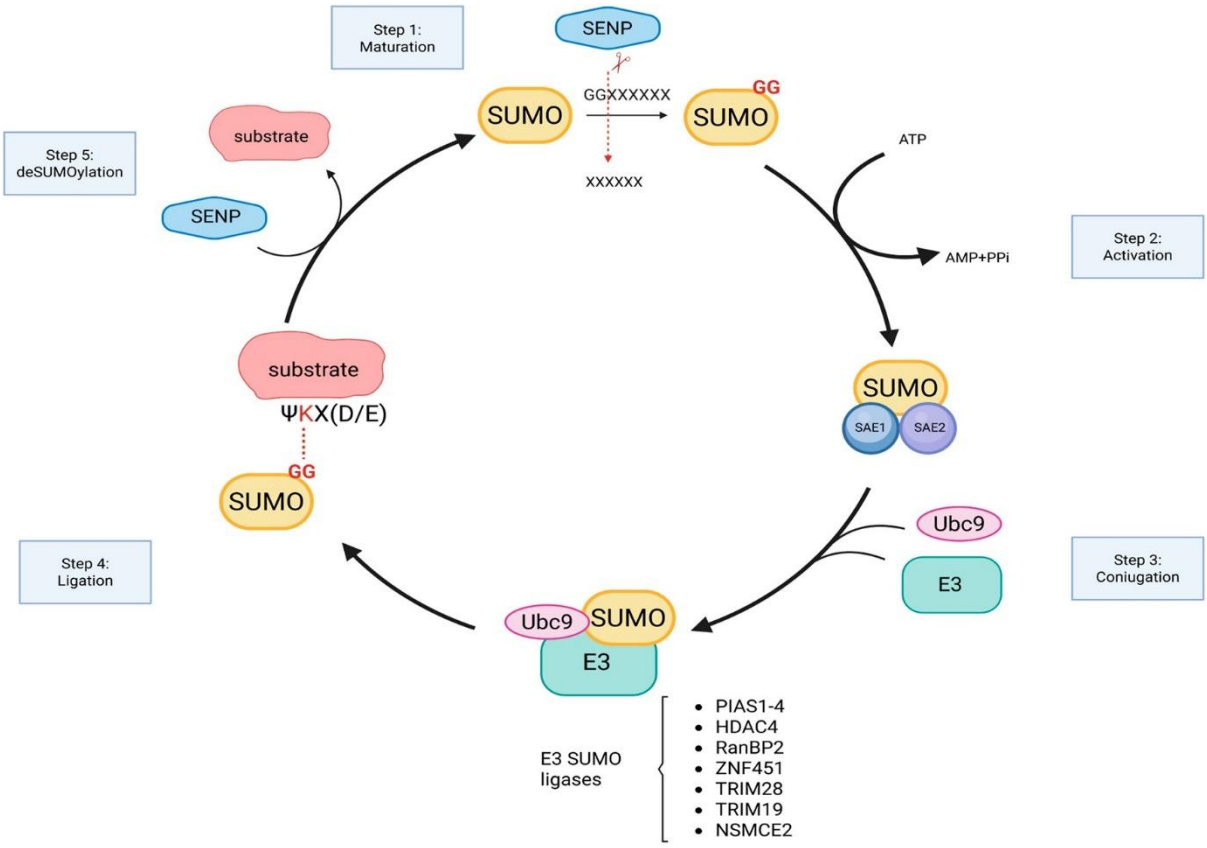


Figure 1.2: Graphical representation of SUMOylation enzymatic cascade (31)

1.3.5 SUMOylation in DNA damage response and replication fork protection

The SUMO signaling pathway is located at the center of the DNA damage response (DDR). SUMOylation affects the assembly of repair foci, influences chromatin's attractive properties for proteins that respond to damage and are involved in repair, and acts in concert with ubiquitin pathways, implying these proteins are either stabilized in the repair pathway or targeted for alteration. (23,29,30,32). PIAS1 and

PIAS4 are E3 SUMO ligases that enhance the signaling of damage and facilitate recruitment of target proteins for repair at double strand breaks, revealing a role for local SUMO complexes in regulating the intensity of the signaling pathways and the intensity of the signal (33). SUMO is also implicated in processes associated with homologous recombination (HR), such as end resection, chromatin remodeling and the recruitment of repair proteins (34).

It is also important at replicated forks that are stressed. How SUMO and ubiquitin synapse to regulate the activity of the replication origins, fork progression and bypass/repair from an impeded fork is revealed by a study. (35). Meanwhile, another study revealed that the SUMOylation catalysed by PIAS4 is required for the activity of CtIP involved in DNA end resection and protection of the replication fork, making the link between SUMO E3 ligase activity and the stability of the replication fork under stress. The role of NSMCE2 in replication stress tolerance is natural as it is part of a complex that binds to some DNA problems, SMC5/6, serving as a local SUMO E3 ligase (23,36,37).

1.3.6 SUMOylation and Breast Cancer

SUMOylation pathway plays a protagonist role in the progression of breast cancer, especially in aggressive TNBC sub-types. This "SUMO addiction" is required for the tumour to survive and undergo epigenetic changes associated with epithelial–mesenchymal transition (EMT) as a condition of enabling him to grow in the face of chronic proteotoxic and replication stress and drug resistance. An important question is regarding cell adaptation upon the loss of specific elements of this network, such as NSMCE2 which is targeted by SUMOs, in this context an inhibitor of the SUMO-activating enzyme (TAK-981) has been developed (38).

During NSMCE2 deficiency, TNBC cells are able to compensate for these deficiencies by utilizing compensatory signaling for their maintenance of chromatin integrity and DNA repair, perhaps via enhancing the conjugation of proteins with UBC9 or through the SUMO-targeted ubiquitin pathway or PIAS-family E3 ligases. As cells with TNBC survive at the stress barrier to replications, it is important to understand these compensatory nodes since tweaks in SUMOylation balance can impact the stability of the chromosomes, as well as their response to platinum agents or PARP inhibitors.(33,37,39)

CHAPTER II

METHODOLOGY

2. METHODOLOGY

2.1 MATERIALS

2.1.1 Cell Culture Reagents

Dulbecco's Modified Eagle Medium (DMEM; Gibco), DMEM/F12 (Gibco) Dulbecco's Phosphate Buffered Saline 1X (PBS; HiMedia), Trypsin (HiMedia), Fetal Bovine Serum (FBS; Gibco), Dimethyl sulfoxide (DMSO), Sodium pyruvate (Gibco), Penicillin-Streptomycin (Gibco), Trypsin (HiMedia)

2.1.2 RNA Extraction Reagents

RNAiso Plus TRIzol reagent (Takara), Chloroform, Isopropanol, 75% Ethanol (prepared in molecular-grade water), Molecular-grade water.

2.1.3 cDNA Synthesis and qPCR Reagents

PrimeScript™ RT Reagent Kit (Takara), TB Green® Premix Ex Taq™ II kit (Takara), molecular grade water; gene-specific forward and reverse primers for NSMCE2, the housekeeping genes 18S rRNA and GAPDH (sequences listed in Table 2.8).

2.1.4 Protein Extraction Reagents

RIPA Lysis Buffer (HiMedia), Protease Inhibitor Cocktail (PIC; 1×, HiMedia), Phenylmethyl sulfonyl fluoride (PMSF; 1 mM, SRL), Bradford Reagent (HiMedia), Bovine Serum Albumin (BSA; for standard curve preparation), Pierce™ BCA Protein Assay Kit (Thermo Fisher Scientific),

2.1.5 SDS-PAGE and Western Blotting Reagents

Distilled water, 30% Acrylamide-bisacrylamide solution, Tris Base (1.5 M, pH 8.8 for resolving gel; 0.5 M, pH 6.8 for stacking gel), 10% Sodium dodecyl sulfate (SDS), 10% Ammonium persulfate (APS), Tetramethylethylenediamine (TEMED), Running buffer (25 mM Tris Base, 192 mM Glycine, 0.1% SDS), 5× Laemmli sample loading dye, Prestained Protein Ladder (HiMedia), Transfer buffer (25 mM Tris Base, 192 mM Glycine, 20% Methanol), 1× PBST (phosphate-buffered saline with 0.05% Tween-20),

Blocking buffer (2% BSA in 1× PBST), Primary antibodies: anti-NSMCE2 (rabbit polyclonal; Protein Tech) and anti-β-actin (mouse monoclonal; Santa Cruz), Secondary antibodies: HRP-conjugated anti-rabbit IgG (NSMCE2) and HRP-conjugated anti-mouse IgG(β-actin), Clarity™ Western ECL Substrate (Bio-Rad).

2.1.6 Miscellaneous Reagents

Crystal violet (0.5%), Paraformaldehyde (3.7%), Colchicine, 0.075 M potassium chloride (KCl; hypotonic solution), fixative solution (methanol: glacial acetic acid, 3:1 v/v), Olaparib (Sigma Aldrich).

2.1.6 Labware

15 mL Conical-bottom centrifuge tube (Abdos Life Sciences), 50 mL Conical-bottom centrifuge tube (Abdos Life Sciences), 35 mm Cell culture dish (Abdos Life Sciences), 6-Well Plate with lid (CytoOne), 96-well plate (Abdos Life Sciences), Microcentrifuge tubes (1.5ml), Nitrocellulose membrane (HiMedia), Blotting sheets (Bio Rad).

2.1.7 Equipment and Instruments

Inverted Microscope (CKX53; Olympus, Japan), 37°C CO_2 Incubator (PHCBi, Singapore), −20°C, −4°C and −80°C refrigerators, Centrifuge (5702 R; Eppendorf, USA), 37°C Rotex water bath, Bio-safety cabinet Level-2 (Alpha Linear), TC20 Automated Cell Counter (Bio Rad), Ice flaker, Micropipettes (10 µL, 20 µL, 200 µL, 1000 µL; Eppendorf, USA), Micropipette tips, PCR Thermocycler (Mastercycler X50a; Eppendorf), RT-qPCR Thermocycler (QuantStudio 5; Thermo Fisher Scientific), Gel Documentation System (ChemiDoc™ XRS+), Dry Bath (BIO-WORLD™), Magnetic stirrer, pH meter, NanoDrop Spectrophotometer (Thermo Fisher Scientific), Multiskan™ Microplate Reader, Mini-PROTEAN Tetra Vertical Electrophoresis Cell (BioRad), PowerPac Universal Power Supply (BioRad), RNA extraction hood (Bio Bee vertical laminar air flow).

2.2 METHODS

2.2.1 CELL CULTURE

a) Cell lines

Three human breast cancer cell lines were used as experimental models for this study: MDA-MB-231 and MDA-MB-468, both representing the triple negative breast cancer (TNBC) subtype, and MCF-7, a luminal A Estrogen receptor-positive (ER+) breast cancer cell line included as a comparative control. All cell lines were obtained from the NUCSER liquid nitrogen (LN_2) facility. MDA-MB-231 and MDA-MB-468 cell lines were maintained in Dulbecco's Modified Eagle Medium (DMEM; Gibco) supplemented with 10% fetal bovine serum (FBS), 1% penicillin-streptomycin and 1mm sodium pyruvate to support optimal growth of cells. MCF-7 cell line was maintained in DMEM-F12 medium, supplemented with 10% FBS, 1% penicillin-streptomycin. Cultures were incubated at 37°C, 5% CO_2 and subcultured upon reaching 80-90% confluency. All experiments described in subsequent sections were performed across all three cell lines unless otherwise specified.

b) Passaging of cells

Cell lines were passaged upon reaching 80-90% confluency in a 35mm culture dish. The spent culture medium was aspirated, and the cell monolayer was gently washed with 1 mL PBS to remove any residual media and serum. Following PBS removal, 200 µL of pre-warmed trypsin-EDTA solution was added to dish and incubated at 37°C for 3 minutes to facilitate detachment of adherent cells. The dish was gently tapped to ensure complete detachment, and then 1 mL of complete DMEM was added to neutralise the trypsin activity. The cell suspension was gently pipetted 2-3 times to achieve a single cell suspension and transferred to a 15 mL conical tube. Cells were centrifuged at 1,200 rpm for 3 minutes to pellet. The supernatant was discarded, and the pellet was washed with 1 mL PBS and centrifuged again under the same

conditions. The resulting pellet was resuspended in fresh complete medium and seeded in volumes as required in 35 mm culture dishes. Cells were incubated at 37°C with 5% CO_2 until they reached 80–90% confluency for use in downstream experiments. The passaging procedure was carried out in the same manner for each of the three cell lines.

2.2.2 PROTEIN ANALYSIS

a) Protein extraction from cells

To extract the protein from all three cell lines, the incubation was continued for 48 hours after the transfection. Each sample was tripled and then individually collected into a microcentrifuge tube. The cell pellets were lysed with ice-cold 200 µL of RIPA lysis buffer (HiMedia) containing 1× PIC and 1 mM PMSF per pellet. To ensure the complete lysis of the cells the cell pellet was homogenized by aspirating it 15-20 times through a 2 mL plastic syringe. This lysate was then centrifuged at 12,000 rpm for 10 minutes at 4℃. The supernatant was carefully transferred to another 1.5 mL microcentrifuge tube after the supernatant was clarified from the precipitated protein, and stored at −20°C until further use.

b) Estimation of protein using Bradford assay and BCA assay.

We used 2 complementary colorimetric assays, the Bradford assay and the Bicinchoninic Acid (BCA) assay to quantify the protein concentration in cell lysates. The two methods used the bovine serum albumin (BSA) as the reference standard for creating a standard curve.

The Bradford assay was used to establish a standard curve for estimation of BSA, which was plotted against a set of known BSA concentrations ranging from 0 to 10 µL (final volume) stored in individual wells of a 96-well microplate. The experimental samples were added 2 µL in duplicate into separate wells and made up to 10 µL with

distilled water as was done for the standards. In each well 200 µL of Bradford reagent was added and the plate was incubated for 15 minutes at room temperature away from any light that may degrade the reagent. After incubation absorbance was measured at 595 nm with a Multiskan™ microplate reader. A standard curve was graphically plotted from the absorbance of the BSA standards and the protein content of the samples was calculated from the slope of this curve. Microsoft Excel was used to obtain the final protein concentrations.

Protein estimation was done by BCA assay as per the manufacturer's instructions with Pierce™ BCA Protein Assay Kit. BSA standards were prepared at concentrations of 0, 25, 125, 250, 500, 750, 1,000, 1,500, and 2,000 µg/mL. Working reagent was prepared following the ratio of 50:1 (v/v) of BCA Reagent A:B, reagent B. 10 µL of standard and protein sample was mixed with 200 µL of working reagent in a 96-well microplate. This plate was kept at room temperature in the dark for 30 minutes. Absorbance was measured at 562nm with microplate reader and concentration of protein was calculated by the curve.

c) SDS-PAGE AND WESTERN BLOTTING

Western blotting was performed to assess NSMCE2 protein expression in MDA-MB-231, MDA-MB-468, and MCF-7 cells. Polyacrylamide gels were prepared according to standard protocols. A 12% resolving gel was prepared as described in Table 2.2, and a 5% stacking gel was prepared as described in Table 2.3.

Table 2.2: Components and volume for preparing resolving gel (12%)

Component	Volume
H_2O	3.3 mL
30% Acrylamide: Bis-acrylamide	4 mL

Component	Volume
Tris (1.5 M, pH 8.8)	2.5 mL
10% SDS	100 µL
10% APS	100 µL
TEMED	4 µL

Table 2.3: Components and volume for preparing stacking gel (5%)

Component	Volume
H_2O	3.4 mL
30% Acrylamide: Bis-acrylamide	830 µL
Tris (1 M, pH 6.8)	630 µL
10% SDS	50 µL
10% APS	50 µL
TEMED	5 µL

The appropriate volume of protein for each cell line was mixed with 5× loading dye in 4:1 ratio and denatured by heating at 100°C for 8 minutes. Samples were briefly centrifuged and loaded into the wells of the SDS-PAGE gel. Electrophoresis was carried out at 60 V for the initial stacking phase (30 minutes), followed by separation at 90 V.

Following electrophoresis, proteins were transferred to a nitrocellulose membrane using the Bio-Rad wet transfer system at 16 V for 16 hours. The membrane was blocked with 2% BSA prepared in 1× PBS containing 0.05% Tween-20 (PBST) for 3+ hours at room temperature on a rocker to prevent non-specific antibody binding. The membrane was then sectioned according to molecular weight markers to separately probe for NSMCE2 and the internal control β-actin. Membrane sections were incubated overnight at 4°C on a rocker with anti-NSMCE2 (4:5000) and anti-β-actin (1:10000) primary antibodies, each diluted in 2% BSA in PBST solution. Membranes were then washed (4–5 times for 15 min/wash) with 1× PBST and then incubated for 1.5–2 h at RT with HRP-conjugated secondary antibodies (anti-rabbit IgG for NSMCE2 and anti-mouse IgG for β-actin, each at 1/10,000 dilution in 2% BSA in 1× PBST). After a last round of PBST washes (8–10 washes for 10 minutes each), the blots were submitted for development employing the Clarity™ Western ECL Substrate (Bio-Rad) and visualised using the Bio-Rad GelDoc imaging system.

2.2.3 RNA ANALYSIS

a) RNA extraction

Total RNA was extracted from MDA-MB-231, MDA-MB-468, and MCF-7 cells using TRIzol reagent. Briefly, cells were trypsinized, collected, and transferred into 1.5 mL microcentrifuge tubes. Cell pellets were lysed by the addition of 1 mL of TRIzol reagent followed by thorough resuspension by pipetting.

On the following day, 200 µL of chloroform was added to each tube per 1 mL of TRIzol used, and the samples were vortexed vigorously for 30 seconds to ensure complete mixing. Mixtures were incubated at room temperature for 15-20 minutes, followed by centrifugation at 12,000 rpm for 15 minutes at 4°C. Following centrifugation, the clear, top aqueous phase containing RNA was carefully transferred to a new microcentrifuge

tube. An equal volume of cold isopropanol was added, and the mixture was gently inverted 15–20 times to precipitate RNA, followed by incubation in -20ºC for 2-3 hours. Samples were subsequently centrifuged at 14,000 rpm for 30 minutes at 4°C to pellet the RNA. The supernatant was discarded carefully. The RNA pellet was washed twice with 75% ethanol, followed by centrifugation at 12,000 rpm for 15 minutes at 4°C each time. After the final wash, the ethanol was removed and the pellet was air-dried for a few minutes. The RNA was dissolved in molecular grade water. RNA concentration and purity were assessed using a NanoDrop spectrophotometer.

b) Complementary DNA (cDNA) synthesis

cDNA was synthesized from total RNA using a reverse transcription kit (PrimeScript™ RT Reagent Kit,Takara). A cDNA synthesis master mix was prepared using the components listed in Table 2.4, and the reaction was assembled in PCR tubes. Typically, 1 µg of total RNA was used per 20 µL reaction volume. The prepared reaction mix was subjected to reverse transcription using a thermal cycler under the conditions outlined in Table 2.5. The synthesized cDNA was stored at −20°C until further use for quantitative real-time PCR.

Table 2.4: Master mix for cDNA synthesis (for 1 reaction)

Components	Volume
5x Buffer	4 µL
RT-enzyme	0.4 µL
Oligod(T)	0.4 µL
Hexamers	0.2 µL

Components	Volume
H_2O	5 µL

Table 2.5: PCR conditions for cDNA synthesis

Step	Temperature	Time
Reverse Transcription	37°C	15 minutes
Inactivation	85°C	10 seconds
Hold	4°C	∞

c) Quantitative Real-Time PCR (RT-qPCR)

Quantitative real-time PCR (RT-qPCR) was carried out using TB Green® Premix Ex Taq™ II kit to assess the relative expression levels of NSMCE2 in MDA-MB-231, MDA-MB-468, and MCF-7 cells. The synthesized cDNA was diluted at a ratio of 1:10 by mixing 20 µL of cDNA with 180 µL of molecular grade water to ensure optimal template concentration.

The PCR master mix was then made up after dilution according to Table 2.6. Reaction mixes consisted of SYBR Green master mix, gene specific forward and reverse primers for NSMCE2, two housekeeping genes (18S rRNA and GAPDH), molecular grade water and diluted cDNA. Technical duplicates with the aim of ensuring consistency and accuracy were performed for each sample. The amplification was achieved with the QuantStudio™ Real-Time PCR System (Applied Biosystems) and optimized standard cycling conditions as shown in Table 2.7. Table 2.8 shows primer

sequences for qPCR. The ΔΔCt method was used to calculate the relative expression level of the genes.

Table 2.6: Master mix for RT-qPCR (for 1 reaction)

Components	Volume
TB Green (SYBR Green)	5 µL
Gene-specific forward primer	0.4 µL
Gene-specific reverse primer	0.4 µL
ROX	0.2 µL
Diluted cDNA	4 µL
Total reaction volume	10 µL

Table 2.7: RT-qPCR cycling conditions

Step	Temperature	Time
1. Initial denaturation	95°C	4mins
2. Denaturation	95°C	10secs
3. Annealing	56°C	30secs
4. Extension	72°C	15secs
5. Melt Curve stage	95°C (15secs) → 60°C (1 min) →95°C (15 secs)	-

Repeated for 40 cycles from step 2 to 4

Table 2.8: Primer sequences used for RT-qPCR

Sl. No.	Oligo Name	Primer Sequence (5'–3')
1.	Human 18s forward primer	GCAGAATCCACGCCAGTACAAG
2.	Human 18s reverse primer	GCTTGTTGTCCAGACCATTGGC
3.	NSMCE2-Set2 forward primer	TCCACAAGATCCAAAGCAACA
4.	NSMCE2-Set2 reverse primer	GGACTCATGACCACAGTCCA
5.	GAPDH forward primer	GGACTCATGACCACAGTCCA
6.	GAPDH reverse primer	TCAGCTCAGGGATGACCTTG

2.2.4 FUNCTIONAL ASSAYS

a) WOUND SCRATCH ASSAY

To assess the effect of Olaparib on the migratory behaviour of MDA-MB-231, MDA-MB-468 and MCF-7 cells a wound healing assay was carried out. Cells in each line were added to individual wells of a 6-well plate, and allowed to grow at 37°C with 5% CO_2 until they grew to a uniform, confluent monolayer. The monolayer was then wounded using a sterile 10 µL pipette tip dragged across the monolayer in a single, straight pull to create a scratch wound. To exclude any free cells and cell debris from the medium, the medium was removed and the wells gently washed with 1mL 1× PBS before fresh complete DMEM medium was placed. The plates were then moved back to incubator and cultured in standard conditions. The movement of migration into the

wound was sampled over time and recorded using a microscope on regular time frames.

Wound Area Quantification

The images taken at each time mark was processed by a custom written, image analysis software program created in-house using Python for automated wound area quantification. The program has been developed using OpenCV and NumPy library for image processing and pandas library for importing and exporting the contents.

The processing pipeline consisted each of the listed operations sequentially: (1) images were first converted to 8-bit grayscale images; (2) edge information was acquired through the application of the gradient operator to draw a line between the cell-free wound region and the rest of the cell-monolayer. (3)The resultant edge image was then thresholded and in order to remove noise a median blur operation was performed (kernel radius 7). (4) The processed binary mask was then processed to find the smallest contour in the image that contained the wound, and the area of the contour was finally entered in square pixels. (5) The program read all the images in a given folder and sequentially ordered them, and automatically retrieved the time points from the filename of the images. (6) The percentage wound closure for each time point was determined by dividing the wound area at that time point by the "time zero" wound area.

Wound Closure (%) = [($Area_{0h}$ − $Area_t$) / $Area_{0h}$] × 100

$Area_{0h}$ is the wound size at time zero, $Area_t$ is the wound size at the respective time point. (7) The resultant images were automatically exported to the Microsoft excel spreadsheet and binary mask images were saved in addition to the processed images for manual checking of wound accuracy. All threshold and filter parameters were adjusted according to the cell line for accurate quantification.

b) CLONOGENIC ASSAY

A clonogenic assay was done to examine the colony formation efficiency of MDA-MB-231, MDA-MB-468, and MCF-7 cells. Cells were harvested from a 35mm dish at approximately 90% confluency by trypsinization and counted using an automated cell counter before seeding. Cells were seeded at the appropriate density into individual wells of a 6-well plate and incubated at 37°C with 5% CO_2 After cells were attached (≈ 2 days), Olaparib was prepared at a working stock concentration of 50 µM. To achieve a final working concentration of 1,000nM, 20 µL of the stock solution was added directly to each well containing complete DMEM. Cells were exposed to olaparib for 4 hours under standard culture conditions, after which the drug-containing medium was aspirated and fresh complete DMEM was then added, and the plates were returned to the incubator for a minimum of 8-10 days to allow colony formation. Once incubation was complete, the medium was removed and gently washed with 1 mL 1x PBS and colonies were fixed with 3.7% paraformaldehyde for 2 hours at room temperature, The fixed colonies were then stained with 0.5% crystal violet solution for 30 minutes, after which the wells were washed 2–3 times with distilled water to remove excess stain and left to air dry. Colonies were quantified.

C) KARYOTYPIC ANALYSIS

Two days were used to perform the karyotyping on MCF-7 cells, with the first day being a mitotic arrest. Colchicine treated cells were harvested on the first day when the cells were about 60% confluent and incubated for 15 hours at 37°C with 40µL colchicine. On the 2nd day, mitotic cells were removed from the spent media and these cells were pooled with the adherent cells obtained by trypsinizing and washing with PBS. This cell suspension was then subjected to treatment with 0.075M KCl at 37°C for 30minutes. The suspension was fixed after the incubation with 5 drops of

fixative. The cells were then spun 2000 rpm for 3 minutes twice in fixative. Lastly it was resuspended in 500 µL fixative and kept in -20°C.

Fixed cell suspension was then placed onto the slide and was tilted then air dried to make metaphase spreads. The next step was to take the air-dried slide and place it under a microscope for further examination at higher magnification.

CHAPTER III

RESULTS

3. RESULTS

3.1 CELL MORPHOLOGY

The three breast cancer cell lines used in this study displayed distinct baseline morphologies that were consistent with their known molecular subtypes.

MCF-7 cells showed a typical epithelial, cobblestone-like appearance and formed tightly packed, dome-shaped colonies with well-defined cell-cell junctions, consistent with their luminal epithelial origin and expression of E-cadherin. Cell growth was relatively uniform, reaching approximately 80% confluence within 48–72 hours of seeding.

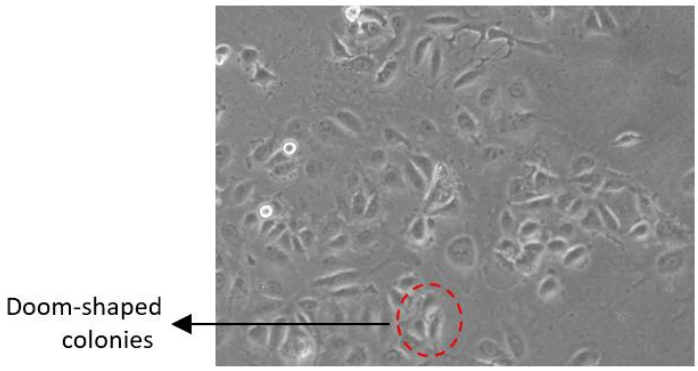


Fig 3.1 b) Morphology of MCF-7 cells under 10X resolution

MDA-MB-231 cells displayed a mesenchymal, unique fibroblast-like morphology with elongated, spindle-shaped cell bodies that are irregular in shape and prominent cytoplasmic projections. The cells grew in a dispersed monolayer with less cell-cell contact compared to MCF-7, reflecting their highly invasive and migratory phenotype. This morphology is consistent with the epithelial-mesenchymal transition (EMT) state of this cell line, which is driven in part by oncogenic KRAS signaling and TP53 mutation.

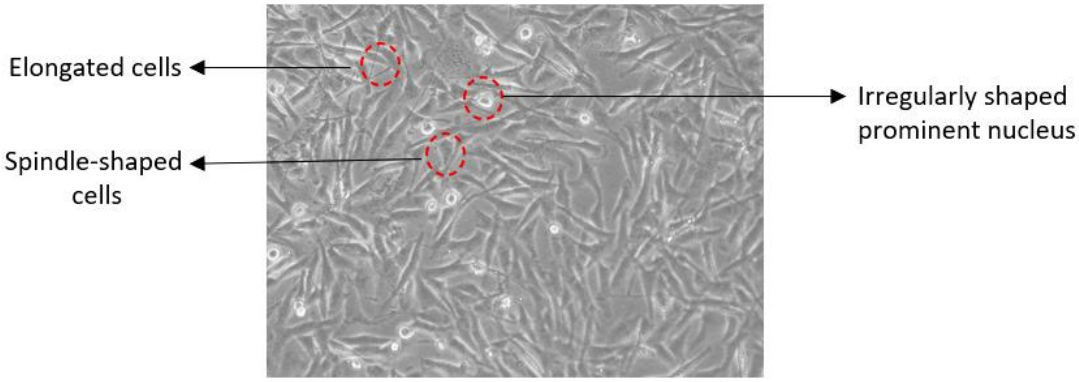


Fig 3.1 b) Morphology of MDA-MB-231 cells under 10X resolution

MDA-MB-468 cells exhibited a rounded morphology and grew in densely packed clusters. These cells displayed a distinctly slower proliferation rate compared to MDA-MB-231. Their compact growth pattern and slower doubling time are consistent with their basal-like transcriptional profile and EGFR amplification-driven biology.

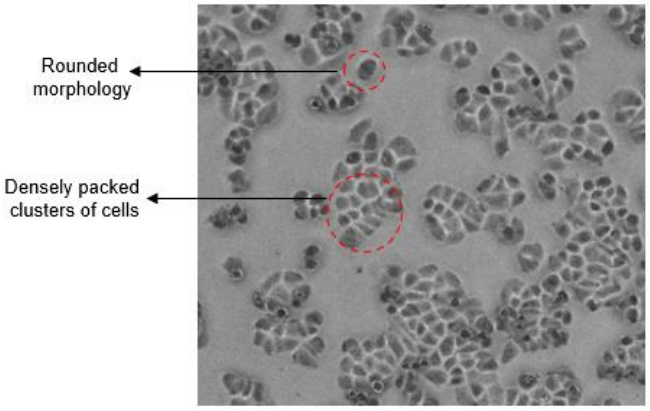


Fig 3.1 c) Morphology of MDA-MB-468 cells under 10X resolution

3.2 RNA EXTRACTION

Total RNA was extracted from MCF-7, MDA-MB-231, and MDA-MB-468 cell lines using TRIzol reagent, following the standard phase-separation protocol. The RNA extraction procedure was extensively optimized to achieve consistent yields of high-quality RNA suitable for downstream quantitative real-time PCR (qPCR) analysis. The optimized protocol consistently yielded A260/A280 ratios of approximately 1.8–2.0 and A260/A230 ratios exceeding 2.0. RNA integrity was confirmed by agarose gel

electrophoresis, which demonstrated distinct, intact 28S and 18S ribosomal RNA bands.

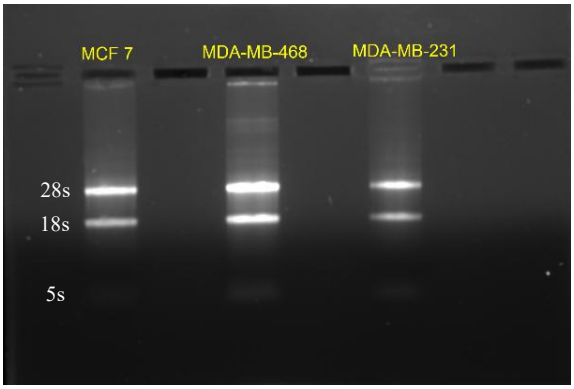


Fig 3.2 Agarose gel electrophoresis of extracted RNA

Table 3.1 Concentration and purity of extracted RNA from the three cell lines

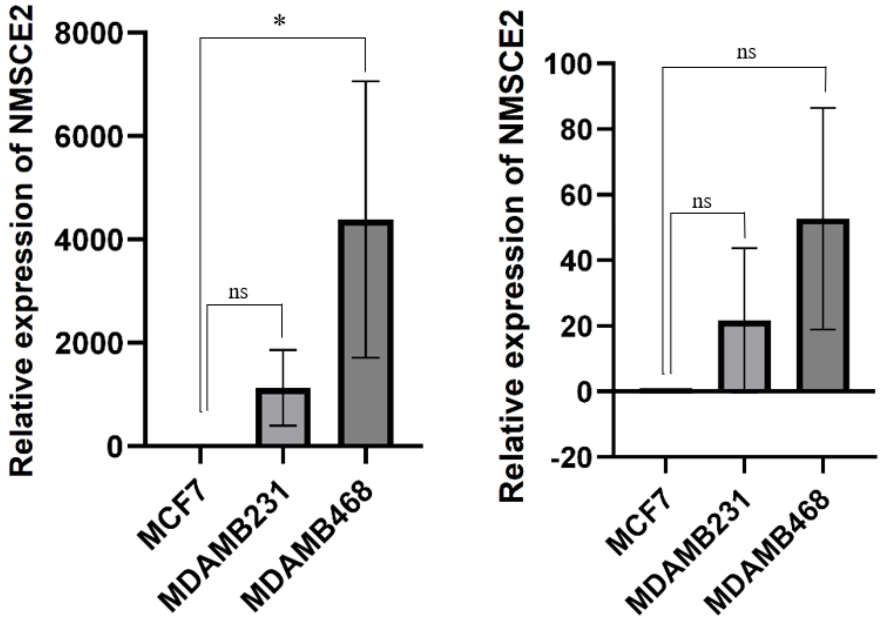
Cell line used	**Concentration of RNA (ng/µL)**	**A_{260}/A_{280} ratio**	**A_{260}/A_{230} ratio**
MCF-7	600	1.97	2.06
MDA-MB-231	93.6	1.71	2.1
MDA-MB-468	206.5	1.8	2.2

3.3 QUANTITATIVE REAL-TIME PCR (RT-qPCR)

To establish the baseline expression of NSMCE2 across the cell lines used in this study, quantitative real-time PCR was performed using GAPDH and 18S rRNA as housekeeping reference genes for normalization. Three independent qPCR experiments were conducted to ensure reproducibility.

Relative quantification (RQ) was performed using the $2^{-\Delta\Delta ct}$ method, with MCF-7 serving as the calibrator (RQ = 1.00). Both normalization methods confirmed that NSMCE2 is expressed at much higher levels in TNBC cells than in the MCF-7 control. There were also differences in fold-change values in different housekeeping genes; however, as fold-changes of 18s and GAPDH were the same, this consistency further

emphasizes the trend and the strong effect of the choice of reference gene on relative quantification. Normalized to 18s rRNA, expression of NSMCE2 was 1,128 folds higher in MDA- MB-231 and 4,387 folds higher in MDA-MB-468 than that in MCF-7. The rise in MDA-MB-468 was significant ($p = 0.047$) and there was a strong upward trend in MDA-MB-231 which just failed to reach significance ($p = 0.056$). These large fold changes most likely are due to their great proliferative activity; in fact, 18s rRNA increase is associated with the greater ribosome biogenesis needed for fast growing cancer cells. Normalization against GAPDH yielded less dramatic but equally significant increases in expression: 52.7-fold in MDA-MB-468 and 21.6-fold in MDA-MB-231. Even though the variance between trials was high in both cases, the biological trend of NSMCE2 gene expression (MDA-MB-468 > MDA-MB-231 > MCF-7) is still unambiguous. The overlap between MCF-7 and MDA-MB-468 expression levels in both normalization strategies suggests that expression of NSMCE2 is consistently enhanced in TNBC cell lines.



a) b)

Fig 3.3 a) Relative gene expression of NSMCE2 using 18s rRNA as control. b) Relative gene expression of NSMCE2 using GAPDH as control

3.4 PROTEIN EXTRACTION

For immunoblotting, all cell lines were lysed with RIPA lysis buffer, and total protein was harvested from cells. One of the biggest problems that were encountered in this study was the accuracy of determining the protein concentration; the Bradford assay was used and later changed to BCA assay.

Bradford assay was used to quantify the initial loading control band intensities for β-actin every lane, which gave inconsistent results. All RIPA buffer components have been identified to interfere with the Coomassie Brilliant Blue dye, which results in systematic underestimation of protein amounts and unequal loading. So, the quantification method was changed to the BCA assay which is very compatible with detergent containing lysis buffer. After this shift, each of the resulting Western blots was characterized by uniformly defined bands of β-actin - loading control for all three cell lines and confirmed that all cells loaded an equal amount of total protein per lane.

Table 3.2 Protein Concentrations from cell lines

Cell line	Concentration of extracted protein(µl/µL)
MCF-7	1.76
MDA-MB-231	1.12
MDA-MB-468	1.36

3.5 WESTERN BLOT ANALYSIS OF NSMCE2 PROTEIN EXPRESSION

Western blotting was performed to validate the qPCR findings at the protein level and to establish the differential expression of NSMCE2 across the cell lines.

The amount of protein samples loaded for SDS-PAGE was extensively optimized. After optimization, the western blot data consistently demonstrated that NSMCE2

protein (~28-29 kDa) is expressed at significantly higher levels in the TNBC cell lines (MDA-MB-231 and MDA-MB-468) compared to MCF-7. The β-actin (~40 kDa) loading control showed uniform, well-defined bands of comparable intensity across all three lanes, confirming equal protein loading. This protein-level data validates the qPCR findings.

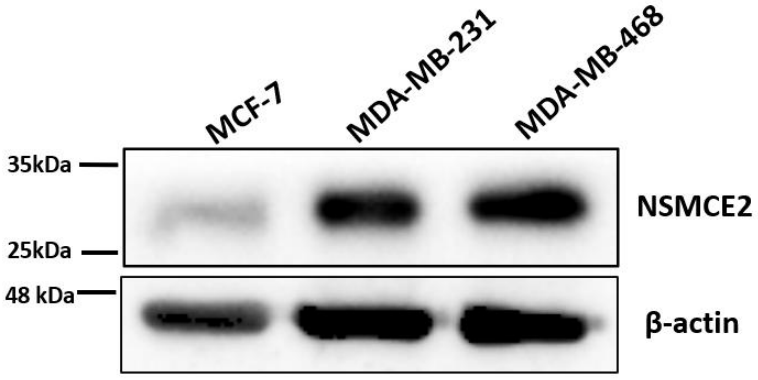


Fig 3.4 Western blot result of NSMCE2 protein expression against β-actin as control

3.6 WOUND SCRATCH ASSAY

In control MCF-7 cells, near-complete wound closure (~99%) was achieved by 32–48 hours. In contrast, Olaparib-treated MCF-7 cells showed a markedly reduced migration rate, reaching only approximately 45–60% closure by 48 hours.

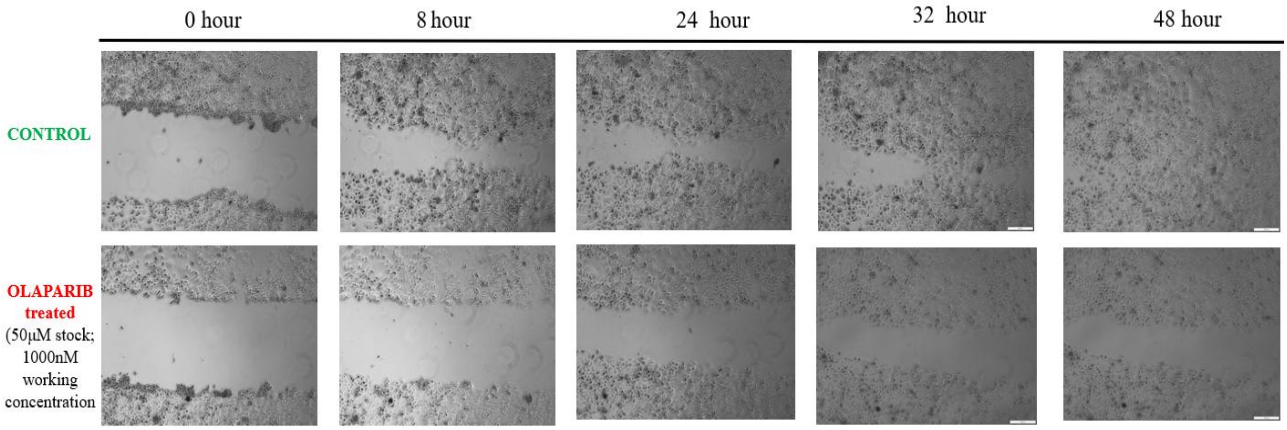


Fig 3.5. Wound healing of MCF-7 cells: Control v/s Treated at different time intervals

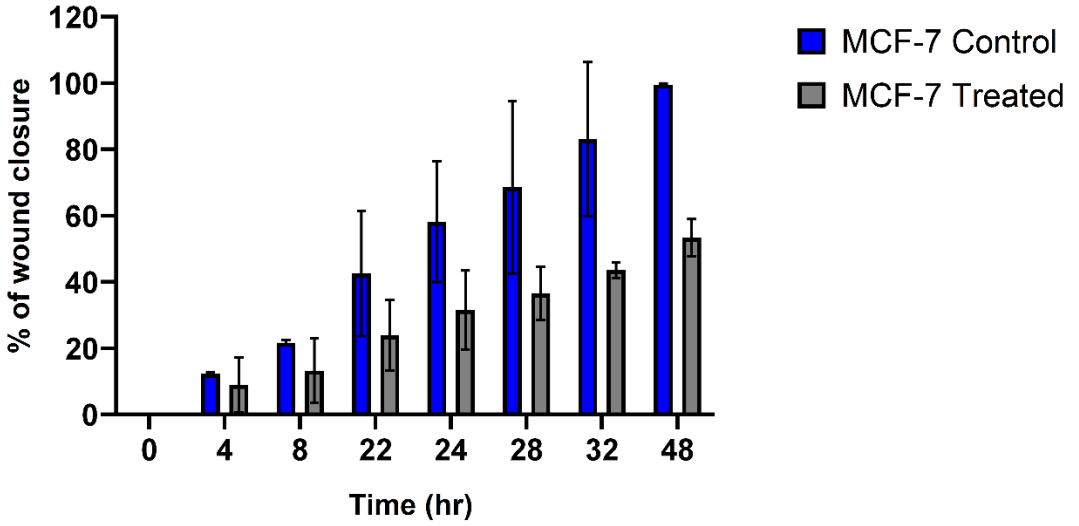


Fig 3.6. Graph representing the percentage wound closure of MCF-7 cells at different time intervals

MDA-MB-231 cells, consistent with their mesenchymal phenotype, demonstrated rapid wound closure, achieving near-complete closure (~93–100%) within 22–24 hours. Olaparib-treated MDA-MB-231 cells also migrated substantially, reaching approximately 87–89% wound closure by 24 hours. The effect of Olaparib on MDA-MB-231 migration was less pronounced than in MCF-7 cells.

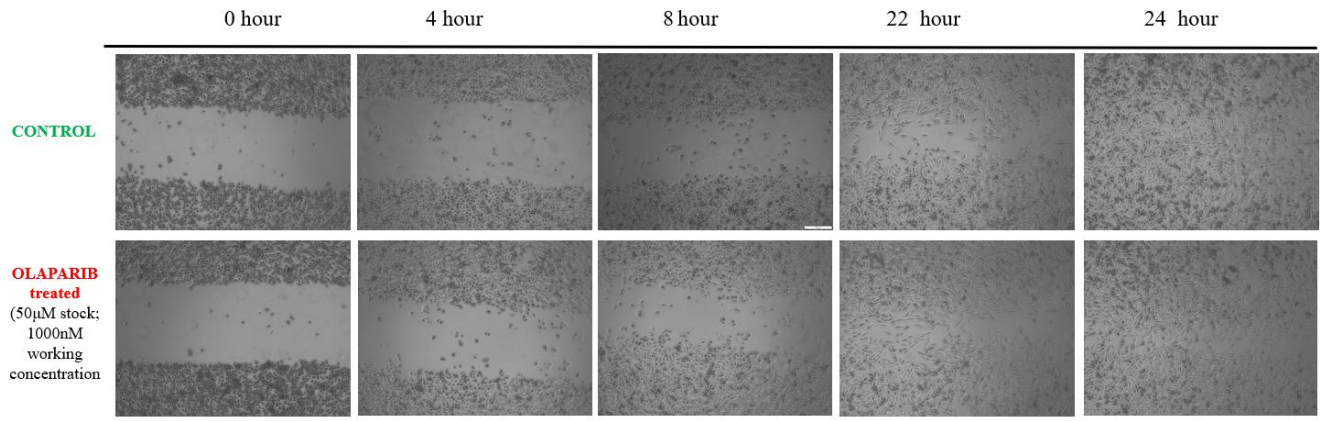


Fig 3.7. Wound healing of MDA-MB-231 cells: Control v/s Treated at different time intervals

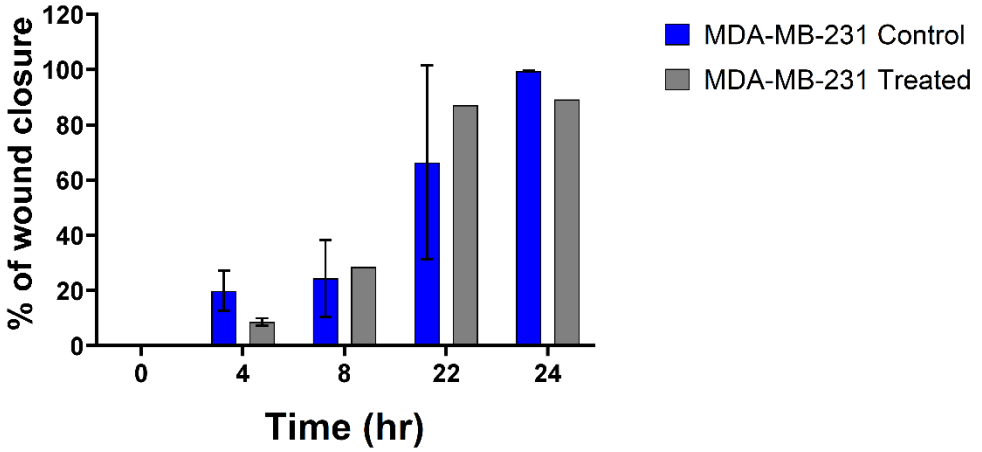


Fig 3.8. Graph representing the percentage wound closure of MDA-MB-231 cells at different time intervals

Control MDA-MB-468 cells reached approximately 85–99% wound closure by 72 hours. Olaparib-treated cells showed a significant delay, reaching only 74–90% closure by 72 hours, with near-complete closure only by 80 hours.

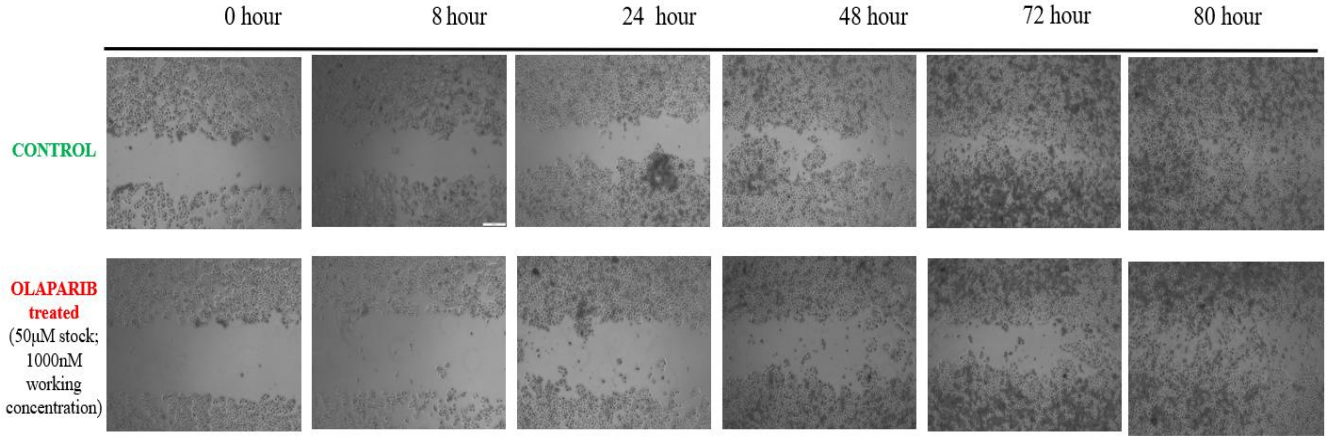


Fig 3.9. Wound healing of MDA-MB-468 cells: Control v/s Treated at different time intervals

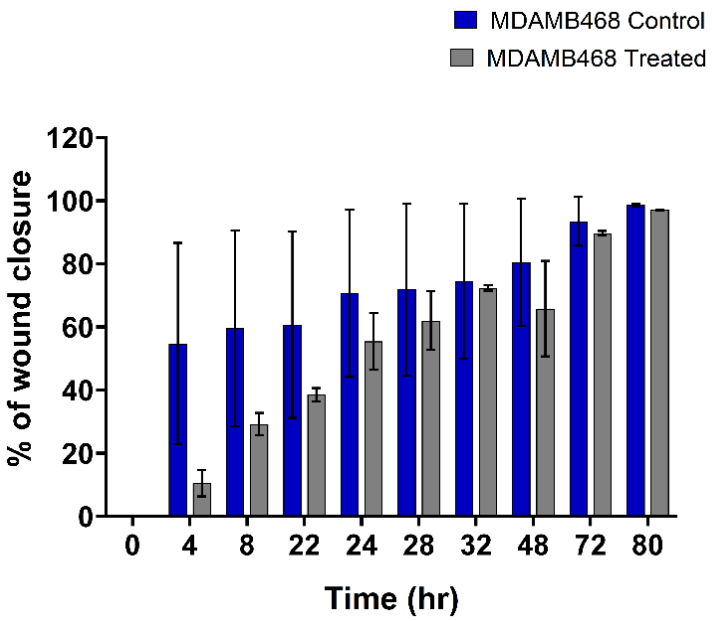


Fig 3.10. Graph representing the percentage wound closure of MDA-MB-468 cells at different time intervals

The wound scratch assay data collectively demonstrated that: (1) MDA-MB-231 cells are the fastest migrating line, consistent with their EMT-driven phenotype; (2) Olaparib visibly inhibits migration across all cell lines, confirming that PARP-dependent DNA repair is important for maintaining migratory capacity; and (3) the TNBC lines, particularly MDA-MB-231, show partial resistance to the anti-migratory effect of Olaparib, which may be linked to their elevated NSMCE2 expression providing compensatory SUMOylation-mediated genomic protection.

While strong trends of inhibition were observed across all three cell lines, the differences did not reach formal statistical significance ($p > 0.05$), likely due to the high variance and small sample size (n=3) typical of these assays.

3.7 CLONOGENIC ASSAY

Clonogenic assay was optimized for MCF-7 and MDA-MB-231 cells; however, MDA-MB-468 cells could not be included in this analysis despite extensive optimization

efforts. The optimal seeding density for the clonogenic assay was determined for each cell line. For MCF-7, a seeding density of 1,500 cells per well was found to produce well-separated, countable colonies. For MDA-MB-231, a lower seeding density of 600 cells per well was optimal.

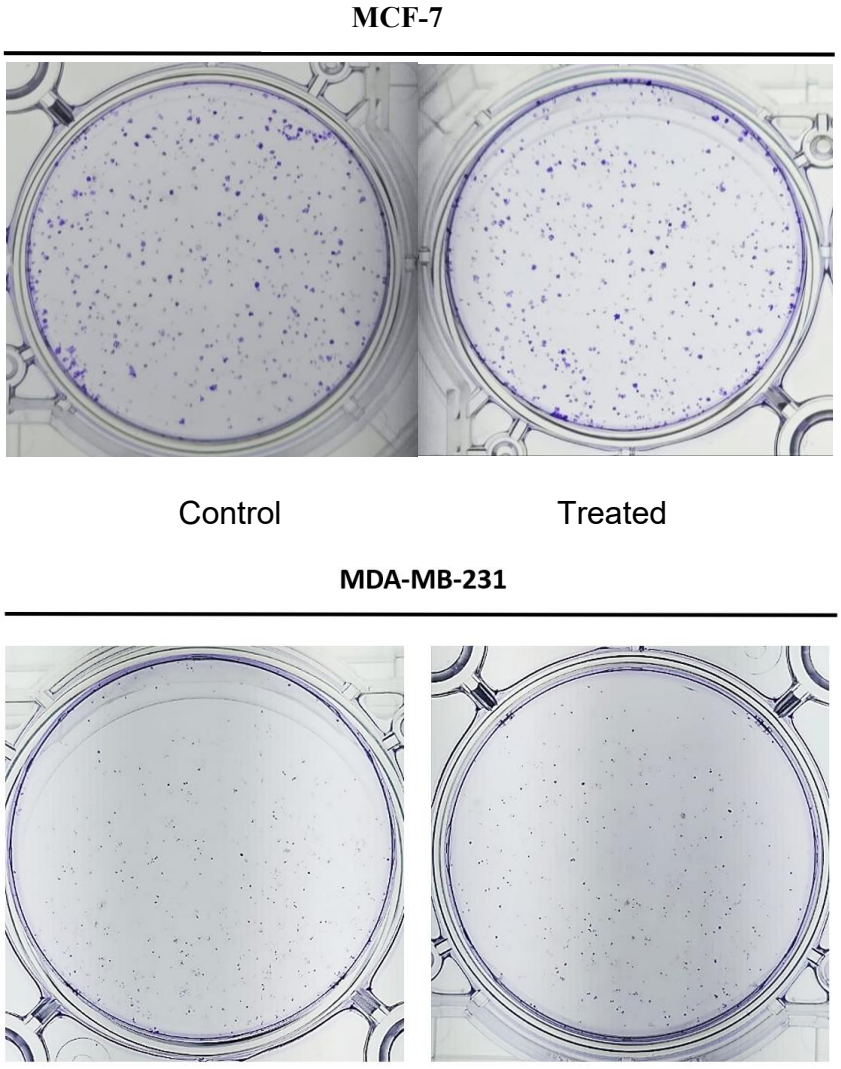


Fig 3.11: Clonogenic survival assay of a) MCF-7 and b) MDA-MB-231 cells.

To evaluate the long-term effect of Olaparib on the proliferative potential of breast cancer cells clonogenic assay was performed. In both MCF-7 (Fig. 3.7 a)) and MDA-MB-231 cell lines (Fig. 3.7 b)), treatment with 1000nM concentration did not result in

a significant decrease in colony numbers compared to the vehicle-treated controls. Both control and treated plates exhibited comparable colony densities and sizes after minimum 10-day incubation period, suggesting that Olaparib at the tested concentration does not inhibit the independent survival or reproductive integrity of these specific cell lines over long term.

3.8 KARYOTYPING

To assess the ploidy and the chromosomal composition of the cell lines used in this study, karyotypic analysis was carried out in MCF-7 and MDA-MB-231 cell lines. There was no analysis conducted of MDA-MB-468 cells. Metaphase spreads of MCF-7 cells revealed that the majority of the cells were hyper triploids. The chromosome counts per spread were always above 60 chromosomes and complex rearrangements were seen at least on one or two occasions, along with the presence of the prominent marker chromosomes. It was possible to observe structural heterogeneity across the population using the condensation quality attained by making use of the colchicine arrest.

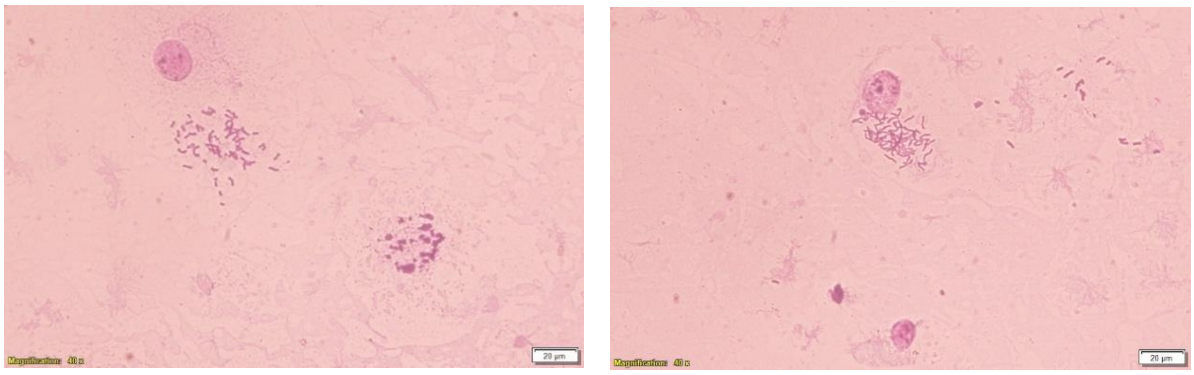


Fig 3.12.a)

The MDA-MB-231 cell line cells also exhibited a modal number that was close to the triploid level. The 231 spreads showed, in a qualitative manner, a higher incidence of

short and very condensed chromosomal elements than did the MCF-7 line. The spreads were clear in all cases and the mode of the number of chromosomes was found in 60-65 with some "stickiness" or clumping noted during the harvest.

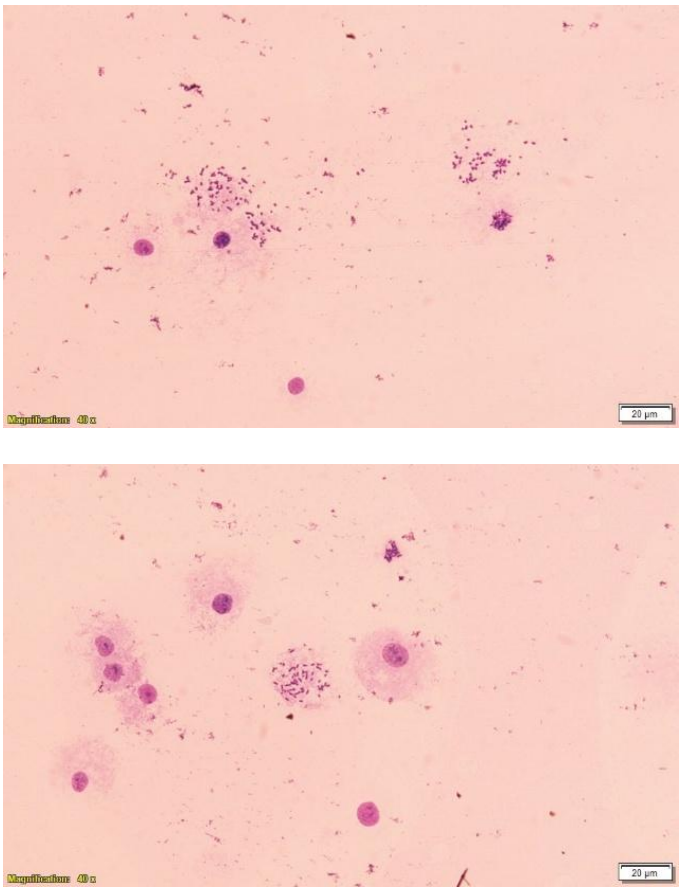


Fig 3.12 b)

Figure 3.12: Comparative Karyotypic Analysis of Breast Cancer Cell Lines: Representative picture of crystal violet stained metaphase spreads (Scale bar = 20 μm) taken at 40x magnification. (A) There are some highly abnormally distributed spreads of the cells with marker-chromosomes. The expression of an expanded spread of (B) MDA-MB-231 cells shows a characteristic near-triploid spread.

CHAPTER IV

DISCUSSION

DISCUSSION

NSMCE2 is an E3 SUMO ligase which forms a part of the SMC5/6 complex. It has proven roles in chromatin stability and repair for stalled replication forks after DNA damage due to replication stress, highlights its relevance as a target for exploration in the field of TNBC. Since TBNC exhibits high, sustained replication stress, argues for genomic instability and cell cycle checkpoint signalling deregulation, it was hypothesized that elevated expression of NSMCE2 could be an adaptive mechanism allowing for these extra cells to withstand the otherwise lethal stress under the conditions of TBNC and to maintain genomic integrity. The results of this study support this hypothesis. Elevated expression of NSMCE2 was discovered in both two TNBC cell lines (MDA-MB-231 and MDA-MB-468) when compared to that of the luminal control cell line MCF-7, suggesting that NSMCE2 upregulation may be associated with the stress-adapted biology found in the aggressive breast cancer subtypes.

On a general level, it is important to say a few words briefly about the baseline morphological description. The low invasion human MCF-7 cell exhibited epithelioid morphology with cuboid shape unlike the others.The human MCF-7 cell showed the typical epithelioid morphology with tight contact with no longer then cuboid shape in accordance with the luminal origin, that also showed a relatively low invasion potential in comparison with the other cells. Due to their high level of invasiveness, the MDA-MB-231 cells' morphology was elongated and appeared to have a disorganized arrangement (scattered). The MDA-MB-468 morphology was more well-rounded, and seemed to form loose groups suggesting a definite basal-like phenotype.

The molecular characterization of the gene expression analysis was done by RT quantitative-PCR, with the purpose of increasing the confidence on the normatization and using 2 reference genes (18S rRNA and GAPDH). In both examples, the NSMCE2 transcript levels are also routinely higher in the TNBC lines, in particular, MDA-MB-468 exhibits the most transcriptional upregulation. That fold-changes depend on which house-keeping gene is used for normalisation is known and, of course, not surprising, because there are fold-changes between the two house-keeping-gene expression (HKE), varying in the cell types. The overall direction and outcome of the normalisation followed the same overall pattern for all of the normalisation methods that were tested; this provided additional assurance that the final result is truly a biological property and not something produced by the experiment. In addition, there was a significant finding of amplifying mRNA levels and comparing the levels of mRNA to protein using a Western blot and a polyacrylamide gel electrophoresis. An important point here is that the expression of either of these two genes is much more than a transcriptional difference and that NSMCE2 is consistently found to be deregulated in TNBC while being likely to be important for the functioning of TNBC cells.

The additional clues of the relationship between higher NSMCE2 and the response to therapeutic drugs were provided from functional assays performed in this study, in TNBC. To test, to a certain extent, if PARP inhibition impacts on cell mobility and cell wound healing the wound scratch assay was carried out. Treatment of the olaparib, the PARP inhibitor resulted in less scratch closure in each cell line used. Indeed, it was important to note that, even after the cells were treated with drugs that could possibly be involved in anti-migratory pathways, the migratory ability of TNBC lines, particularly of MDA-MB-231 was significantly greater in comparison with the MCF-7, suggesting some anti-migratory pathway drug resistance of the TNBC cells. This

partial resistance in the TNBC environment might have been the general adaptability of these cells, whereas, high NSMCE2 might have a role to play in this by enabling their ability to undergo replication induced repair in drug stress. The clonogenic survival experiments also showed no effect on the long term cell growth when different levels of concentration of olaparib were utilised. Conditions had to be optimized for each of the cell lines tested and under the conditions tested, it was determined that the MDA-MB-468 cells would not form large enough colonies to allow a comprehensive comparison study to be conducted with the other cell lines. A problem with the current experimental model, but there is known variation in TNBC and the message here is that even though the results from one experimental model to another means not the same.

Collectively the results of this study strongly support that TNBC cells possess an intrinsic capacity to cope with therapeutic stress and that NSMCE2 could be part of a wider network of DNA damage response pathways that invade in response to therapeutic stress to maintain cell survival. Several deficiencies of the present and present invention, however, in that respect should be mentioned. The data shows that an increase in NSMCE2 has been associated with aggressive TNBC behaviour, however this has not yet been proven to be a direct cause of the more aggressive behaviour. For example, the wound healing assay is used to measure migration as well as proliferation, but cannot discriminate between these two. In addition, the SUMO pathway proteins (PIAS1, PIAS4, UBC9) or the SUMOylated proteins were not targeted in the experimental investment in the present article.Otherwise, this experimental work was limited to a number of biological replicates, and the components of the SUMO pathway (PIAS1, PIAS4, UBC9) and the SUMOylated target proteins could not be directly investigated.

After that, the proliferation, migration and clonogenic survival should be tested again with and without the chemotherapeutic drugs and mRNA and protein confirmation of what would be done after news should a NSMCE2 knock down be performed in the same TNBC cell lines. It would provide another perspective of the Phenotype because of NSMCE2. Overall, these results provide valuable support that NSMCE2 is, at least, a biologically relevant component of the adaptive repair machinery in TNBC and thus should be further explored as a therapeutic target or as a biomarker of treatment resistance.

CHAPTER IV

SUMMARY, CONCLUSION AND SCOPE FOR FUTURE WORK

SUMMARY AND CONCLUSION

In this dissertation molecular and functional assays were used to elucidate the role of NSMCE2 in TNBCs. In both normalisation methods and in both detection methods, NSMCE2 was found to be invariably and consistently elevated in both cell types (MDA-MB-231 and MDA-MB-468) when compared to the levels detected in MCF-7. Although there was a decrease in wound closure in all three lines, functional analysis showed that there was still more mobility in the TNBC cells compared to the MCF-7 cells after the olaparib treatment, particularly the MDA-MB-231 cells. All these together indicate that increased NSMCE2 might be associated with the stress-adapted phenotype of TNBC and with the flexibility of genome stability/repair in aggressive breast cancer. The dissertation does not directly show compensation SUMOylation nor establish a causal relationship between NSMCE2 and the functional phenotypes observed. The study, however, provided a sound experimental basis for future mechanistic studies and paved the way towards an understanding of NSMCE2 as a potential therapeutic target in replication-stress-induced tumours.

SCOPE FOR FUTURE WORK

Future studies should be done using validated NSMCE2 siRNA knockdown TNBC cells used here and begin by evaluating their effects on these cells. The effect of NSMCE2 silencing on proliferation, migration, colony formation and viability should be determined in both the presence and absence of chemotherapeutic agents after the silencing has been optimized. Next, mechanistic studies should be conducted to establish whether the levels of PIAS1, PIAS4, UBC9 and SUMOylated repair substrates are altered as a result of the loss of NSMCE2. Rescue experiments and more biological replicates and drug response testing would be helpful to directly investigate the functional role of NSMCE2 in TNBC and to aid in the translation of this work.

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ANNEXURES

ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NSMCE2-DEFICIENT BREAST CANCER CELLS

A Synopsis Submitted By

Ms. Manohari M Kulkarni

(US No: NU24BSCB03)

To

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1. LIST OF ABBREVIATIONS

Abbreviation	Full Form
TNBC	Triple-Negative Breast Cancer
ER	Estrogen Receptor
PR	Progesterone Receptor
HER2	Human Epidermal Growth Factor Receptor 2
EMT	Epithelial-Mesenchymal Transition
SMC	Structural Maintenance of Chromosomes
NSMCE2	Non-SMC Element 2 (also known as MMS21)
PTM	Post-Translational Modification
DDR	DNA Damage Response
SUMO	Small Ubiquitin-like Modifier
DMEM	Dulbecco's Modified Eagle Medium
FBS	Fetal Bovine Serum
siRNA	Small Interfering RNA
SDS-PAGE	Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis
qRT-PCR	Quantitative Real-Time Polymerase Chain Reaction
PBS	Phosphate-Buffered Saline
DMSO	Dimethyl Sulfoxide
PARP	Poly (ADP-ribose) polymerase
HRR	Homologous Recombination Repair

2. BACKGROUND

Breast cancer remains the most prevalent cancer in women worldwide. While advancements in early detection and treatment have improved patient outcomes in certain subtypes, aggressive forms of breast cancer such as triple negative breast cancer (TNBC) continues to pose significant treatment challenges. TNBC is characterized by the absence of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2). The absence of these receptors is associated with a poor prognosis, high rates of relapse and limited treatment options. The lack of molecular targets leaves chemotherapy as the main treatment option for TNBC. (1)

TNBC is clinically recognized for its aggressive nature, characterized by rapid tumor progression, early recurrence, and a high probability for distant metastases, particularly to the brain and lungs.(2) The aggressiveness of TNBC is caused by its molecular heterogeneity and high genomic instability, with basal like tumors representing the major subtype and frequently exhibiting mutations in TP53, BRCA1/2 or features exhibiting consistent 'BRCAness' phenotype.(1) BRCA2, especially, plays a crucial role in protecting stalled replication forks from nucleolytic degradation by facilitating the recruitment of RAD51 to the fork, thereby stabilizing the replication machinery and promoting fork restart. In BRCA2-deficient cells, hindered replication forks undergo reversal, forming four-way junctions. These reversed forks are highly susceptible to degradation by nucleases such as MRE11, contributing to genomic instability and sensitivity to replication stress. Importantly, blocking fork reversal prevents this degradation, thereby restoring fork integrity in BRCA2-deficient cells (3) Furthermore, other studies suggest that tumor cells can acquire chemoresistance by stabilizing replication forks independently of BRCA2. Loss of factors such as PTIP prevents MRE11 recruitment to reversed forks, stabilizing them and enabling tumor cells to survive DNA-damaging chemotherapy agents like platinum compounds and PARP inhibitors (4). Despite initial responses to chemotherapy, the long-term management of TNBC and high rate of disease relapse remains a challenge due to rapid development of chemoresistance. Current treatment for TNBC primarily relies on conventional chemotherapy, including anthracyclines, taxanes and platinum-based agents like cisplatin, but these fail to achieve effective responses. In recent years, Poly (ADP-ribose) polymerase (PARP) inhibitors have emerged as a promising targeted therapy, particularly for TNBC patients with BRCA1/2 mutations or BRCAness features. PARP

inhibitors exploit the concept of synthetic lethality by blocking DNA repair in tumor cells already compromised in homologous recombination repair (HRR), thereby inducing cell death. In the Indian clinical context, PARP inhibitors such as Olaparib and Talazoparib have shown efficacy in BRCA-mutated metastatic breast cancer, offering a new avenue for personalized treatment. However, their accessibility and routine use in India remain limited due to factors like high treatment costs, lack of widespread BRCA testing, and inconsistent awareness among clinicians. Therefore, identifying reliable molecular predictors of response to PARP inhibitors and improving their accessibility are critical for optimizing TNBC management in India.(5)

Chemoresistance in TNBC can arise through multiple mechanisms, each contributing to the tumor's ability to evade the cytotoxic effects of chemotherapeutic agents. (6) Among the key drivers of chemoresistance in TNBC are defects and adaptations in DNA repair pathways, including replication fork stabilization mechanisms that enable tumor cells to protect delayed or reversed forks from degradation, even in the absence of functional BRCA2 (3). Enhanced DNA repair capacity enables cancer cells to effectively repair chemotherapy-induced DNA damage allowing them to evade cytotoxic effects. A key component of this repair machinery is the Structural Maintenance of Chromosomes 5/6 (SMC5/6) complex, which ensures genomic stability, particularly under replication stress. Within this complex, NSMCE2, a SUMO E3 ligase, provides an important functional link between structural genome maintenance and the regulation of DNA repair through SUMOylation, a post-translational modification (PTM) known to regulate the activity of numerous DNA repair proteins (7). The regulation of protein function within the DNA damage response (DDR) extends beyond the presence of core repair factors and is also controlled by PTMs that dynamically influence protein stability, cellular localization, interaction networks, and enzymatic activity. SUMOylation, among other PTMs, has emerged as a key regulatory mechanism governing genome stability and DDR efficiency. SUMO modifications on DNA repair proteins coordinate the recruitment and retention of repair factors at stalled replication forks, promoting fork stabilization and preventing fork collapse. This SUMO-dependent regulation ensures efficient replication restart and limits genomic instability under stress conditions(8). NSMCE2-mediated SUMOylation within the SMC5/6 complex facilitates efficient DNA repair under stress, protecting genomic integrity. Dysregulation of NSMCE2 or other SUMOylation enzymes effects repair processes, contributing to genomic instability. Emerging studies show that aberrant SUMOylation not only enhances tumor cell proliferation, epithelial–mesenchymal transition (EMT), migration, and therapy resistance, but in certain cases, it can also stabilize tumor suppressors and inhibit

oncogenic pathways (9,10). However, the role of SUMOylation in aggressive breast cancer subtypes like TNBC remains incompletely understood. Furthermore, the potential compensatory SUMOylation mechanisms that may arise in response to disrupted SUMOylation regulation, such as through NSMCE2 deficiency, are yet to be explored in TNBC, representing a significant gap in understanding the molecular basis of chemoresistance and identifying novel targets to improve the treatment results.

3. LITERATURE REVIEW

The interplay between post-translational modifications (PTMs) and genome stability has emerged as a critical area of research in understanding tumor development, progression, and therapeutic resistance. SUMOylation, the covalent attachment of Small Ubiquitin-like Modifier (SUMO) proteins to target substrates, regulates a wide range of cellular processes, including DNA damage repair, chromatin remodeling, transcription regulation, and cell cycle progression. In recent years, alterations in SUMOylation dynamics have been increasingly recognized as contributors to breast cancer pathogenesis, particularly in aggressive subtypes such as triple-negative breast cancer (TNBC). TNBCs are often characterized by elevated SUMOylation activity, as reflected by the overexpression of core pathway components like UBC9 and SAE1, which correlate with enhanced tumor aggressiveness and poor prognosis(11,12). SUMOylation modifies key players involved in DNA damage repair, epithelial-mesenchymal transition (EMT), and transcriptional regulation, thereby influencing tumor progression and therapy response(13,14). Notably, SUMOylation is essential for maintaining the basal-like breast cancer phenotype, a molecular category that significantly overlaps with TNBC (15)

The structural maintenance of chromosomes (SMC) 5/6 complex plays an essential role in preserving genome integrity, with its SUMO E3 ligase subunit, NSMCE2 (also known as MMS21), being a critical contributor to this process. NSMCE2 facilitates DNA damage repair, replication fork stability, and resolution of recombination intermediates, thereby safeguarding chromosomal stability (7). Notably, NSMCE2 has been identified as a tumor suppressor, as demonstrated by Jacome et al. (2015)(16), who reported that complete loss of NSMCE2 in mice models results in severe genomic instability, cancer predisposition, and accelerated aging. Interestingly, this tumor-suppressive role was found to be largely independent of NSMCE2's SUMO ligase activity, indicating that structural or scaffolding functions of NSMCE2 within the SMC5/6 complex also contribute to genome stability.

While NSMCE2's SUMOylation function may not solely account for its tumor-suppressive role, SUMOylation remains a critical pathway for mitigating genotoxic stress, particularly in the context of NSMCE2 deficiency. Cancer cells, though benefiting from moderate genomic instability to drive evolution and progression, face a threshold beyond which excessive DNA damage compromises viability. It is within this crucial balance that compensatory SUMOylation mechanisms may emerge as a survival strategy. Extensive research has shown that SUMOylation of key DNA repair and replication stress response proteins, including BRCA1, 53BP1, and CtIP, enhances genome maintenance under stress conditions (17,18). Specifically, SUMOylation of CtIP, a central regulator of DNA end resection, promotes homologous recombination and facilitates efficient DNA repair following double-strand breaks (19). Furthermore, global SUMO pathway reprogramming has been shown to occur in cancer cells exposed to replication stress, acting as a protective mechanism to stabilize stalled replication forks and ensure cell survival(20).

Thus, in the context of NSMCE2 deficiency, cancer cells experience heightened DNA damage and replication stress(16). To prevent lethal genomic instability-especially under chemotherapy-they may activate compensatory SUMOylation pathways, upregulating PIAS1/4 or similar ligases to preserve DNA repair activity and replication fork stability. While direct experimental evidence for this compensatory SUMOylation response in NSMCE2-deficient cancer cells remains to be established, the mechanistic foundation linking NSMCE2 loss, genome instability, and SUMO-dependent stress adaptation provides a strong rationale to explore this hypothesis further.

4. RESEARCH QUESTION:

Does NSMCE2 deficiency in triple-negative breast cancer (TNBC) cells lead to a compensatory upregulation and altered activity of other E3 SUMO ligases, and if so, does this compensatory SUMOylation enhance adaptive DNA repair and promote cell survival under cytotoxic stress?

5. AIM AND OBJECTIVES

Aim: To investigate the role of compensatory SUMOylation mechanisms in maintaining DNA repair capacity and promoting chemoresistance in NSMCE2-deficient triple-negative breast cancer (TNBC) cells.

Objectives:

1. To downregulate NSMCE2 expression in MDA-MB-231 cells using siRNA transfection and confirm knockdown efficiency.
2. To investigate the effects of NSMCE2 downregulation on key cellular processes.

6. METHODOLOGY

Objective 1: *To downregulate NSMCE2 expression in TNBC (MDA-MB-231) cells using siRNA transfection and confirm knockdown efficiency.*

Task (1): To determine the baseline transcript and protein-level expression of SUMOylation pathway proteins and known interacting proteins.

MDA-MB-231 cells will be cultured in DMEM supplemented with 10% fetal bovine serum (FBS) and 1% Penicillin-Streptomycin at 37°C in a 5% CO_2 incubator. Prior to transfection, total RNA will be isolated using TRIzol reagent according to the manufacturer's instructions. RNA purity and concentration will be assessed using a spectrophotometer. First-strand cDNA will be synthesized from the isolated RNA using a commercial reverse transcription kit. qRT-PCR will be carried out using gene-specific primers for target transcripts such as NSMCE2, SUMO1, SUMO2/3, UBC9, BRCA1, and BRCA2. 18S rRNA shall be used as an internal housekeeping gene. Relative transcript expression levels will be calculated.

For protein analysis, whole-cell extracts will be prepared, and protein concentration will be measured using the Bradford assay. Equal amounts of protein (40–50 µg) will be separated on SDS-PAGE, transferred onto nitrocellulose membranes, and Western blotting will be performed using antibodies specific to NSMCE2, SUMO pathway proteins, BRCA1, BRCA2, and Beta-actin as loading control.

Task (2): siRNA transfection-based downregulation of NSMCE2 in MDA-MB-231 cells.

MDA-MB-231 cells will be transfected with NSMCE2-specific siRNA using Lipofectamine RNAiMax following an optimized protocol. Scrambled siRNA will be used as a negative control. After 48 hours of transfection, both siRNA-transfected and control cells will be harvested.

Total RNA will be isolated, cDNA synthesis will be performed, and qRT-PCR will be carried out to confirm NSMCE2 knockdown as described above.

Whole-cell protein extracts will be prepared, and protein concentration will be measured. Western blotting will be performed using an anti-NSMCE2 antibody to confirm knockdown efficiency, with Beta-actin serving as loading control.

Task (3): Post-knockdown analysis of transcript and protein-level expression of SUMOylation pathway proteins and known interacting proteins.

Following NSMCE2 downregulation, expression levels of SUMO1, SUMO2/3, UBC9, BRCA1, and BRCA2 will once again be assessed at the transcript and protein levels using qRT-PCR and Western blotting as described previously to evaluate compensatory changes.

Objective 2: *To investigate the effects of NSMCE2 downregulation on key cellular processes such as viability, proliferation, and migration in TNBC cells.*

Task (1): Clonogenic assay to evaluate long-term proliferation potential.

siRNA-transfected and control MDA-MB-231 cells will be seeded at low density in culture plates and allowed to grow for 10-14 days to form colonies. The seeding density will be optimized to avoid overlapping colonies. After incubation, the colonies will be fixed with methanol-formalin and stained with crystal violet. The number of colonies will be counted under a microscope, and the survival fraction will be calculated to assess the impact of NSMCE2 downregulation on colony formation and proliferation.

For all assays, four experimental conditions will be maintained:

- Control cells (untreated)
- Control cells + chemotherapeutic drug (e.g., platinum-based drug, doxorubicin, or PARPi)
- NSMCE2 knockdown cells (untreated)
- NSMCE2 knockdown cells + chemotherapeutic drug

Task (2): MTT assay to assess short-term cell viability.

siRNA-transfected and control cells will be seeded in a 96-well plate at an appropriate density to ensure exponential growth during the experiment. After incubation for 24, 48, and 72 hours, MTT reagent (0.5 mg/ml) will be added to each well and incubated for 2-4 hours. Formazan crystals formed by viable cells will be dissolved using dimethyl sulfoxide (DMSO), and absorbance will be measured at 570 nm using a microplate reader to determine cell viability.

Task (3): Wound scratch assay to assess cell migration.

siRNA-transfected and control cells will be seeded in 6-well plates and grown to form a uniform monolayer. A scratch will be made across the cell monolayer using a sterile pipette tip. Detached cells will be removed by washing with PBS, and fresh medium will be added. Images of the scratch area will be captured at 0, 12, 24, and 48 hours using a microscope. The percentage of wound closure will be quantified to evaluate the effect of NSMCE2 downregulation on cell migration.

Task (4): Metaphase spread preparation and chromosomal aberration analysis.

After 48–72 hours of transfection, siRNA-transfected and control MDA-MB-231 cells will be treated with Colcemid to arrest them in metaphase. Cells will be harvested and subjected to hypotonic treatment using 0.075 M KCl, followed by fixation with methanol: acetic acid (3:1). Metaphase spreads will be prepared by dropping the fixed cells onto clean glass slides. After air drying, the slides will be stained with Giemsa stain and observed under a microscope. The number and type of chromosomal aberrations, including breaks, gaps, and aneuploidy, will be analysed to assess genomic instability resulting from NSMCE2 downregulation.

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8. BUDGET REQUIREMENTS:

This project will require maximum of 50000 INR

9. ETHICAL CONSIDERATION:

The proposal has minimal biosafety issue. It does not involve use of any samples from humans. All experiments using cell lines will be carried out in a cell culture facility with proper protocols in place pertaining to discard of materials.

10. JUSTIFICATION/IMPLICATION

This study aims to explore these compensatory SUMOylation pathways in NSMCE2-deficient TNBC cells and assess how they modulate sensitivity to chemotherapeutic agents. Understanding this interplay can provide crucial insights into the mechanisms of drug resistance and identify potential problems that can be therapeutically targeted. The findings from this study may have significant translational implications. Targeting compensatory SUMOylation mechanisms, either alone or in combination with conventional chemotherapy, could enhance therapeutic efficacy and potentially overcome chemoresistance in TNBC. Moreover, this work contributes to the broader understanding of SUMOylation dynamics in cancer biology and offers novel perspectives on manipulating DNA repair pathways for improved cancer treatment outcomes.

11. WORK PLAN

Activity	August-September 2025	October – December 2025	January- Mar 2026	Apr - May 2026
Literature review, cell culture training				
Experimental work for Objective 1: NSMCE2 knockdown and expression analysis (baseline, transfection, post-knockdown qRT-PCR & Western blotting)				
Experimental work for Objective 2: Functional assays i.e., Clonogenic assay, MTT assay, wound-scratch assay, Metaphase spread preparation and chromosomal aberration analysis				
Data analysis, dissertation writing & presentation preparation				

PROTOCOL FOR REGISTRATION OF M.Sc. SYNOPSIS OF DISSERTATION

1	NAME OF CANDIDATE AND UNIVERSITY SEAT NUMBER	MANOHARI M KULKARNI NU24BSCB03
2	NAME OF INSTITUTION AND ADDRESS	NITTE UNIVERSITY CENTRE FOR SCIENCE EDUCATION AND RESEARCH (NUCSER) PANEER CAMPUS, DERALAKATTE, MANGALURU-575018
3	PROGRAM OF STUDY AND COURSE	MASTER OF SCIENCE CANCER BIOLOGY
4	MONTH AND YEAR OF JOINING THE PROGRAM	AUGUST-2024
5	TITLE OF THE TOPIC	ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NCMCE2-DEFICIENT BREAST CANCER CELLS
6	DOES THE STUDY REQUIRE ANY INVESTIGATIONS OR INTERVENTIONS TO BE CONDUCTED ON PATIENTS / HUMANS /ANIMALS IF SO, PLEASE DESCRIBE BRIEFLY	NO

7	ETHICS COMMITTEE CLEARANCE BEEN OBTAINED FROM YOUR INSTITUTION?	**
8	SIGNATURE OF THE CANDIDATE WITH DATE	30/7/25
9	REMARKS OF THE CO-GUIDE	Satisfactory
10	NAME AND DESIGNATION OF CO-GUIDE	DR. SUDARSHAN KINI ASSOCIATE PROFESSOR - (G-1) DEPARTMENT OF BIO & NANOTECHNOLOGY
11	SIGNATURE OF THE CO-GUIDE WITH DATE	11/08/2025
12	REMARKS OF THE GUIDE	Satisfactory

13	NAME AND DESIGNATION OF THE GUIDE	DR. VIVEK TRIPATHI ASSOCIATE PROFESSOR – (G-1) DEPARTMENT OF MOLECULAR GENETICS AND CANCER
14	SIGNATURE OF GUIDE WITH DATE	Vivek Tripathi 30/07/2025

15	NAME OF THE HEAD OF THE DEPARTMENT	Dr. GUNIMALA CHAKRABORTY
16	SIGNATURE OF THE HEAD OF THE DEPARTMENT WITH DATE	G. Gunimala Dei 30/1/25
17	REMARKS OF DIRECTOR/DEAN	Recommended.
18	NAME AND SIGNATURE OF DIRECTOR/DEAN WITH DATE	Prof. DR. ANIRBAN CHAKRABORTY

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Communication of Decision of the Institutional Scientific Review Committee (SRC)

SRC No: M.Sc./CB/2025-61

Project title:	**INVESTIGATING THE ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NSMCE2-DEFICIENT BREAST CANCER CELLS MODULATING CHEMOTHERAPEUTIC RESPONSE**
Student:	**Ms. MANOHARI M KULKARNI [NU24BSCB03]**
Name & Address of Institution:	Nitte University Centre for Science Education and Research, Mangalore-18
New review	~~Revised review~~ ~~Expedited review~~
Date of review (D/M/Y): **26.07.2025** Date of previous review, if revised application: NA	
Decision of the SRC: ~~Recommended~~ **Recommended with suggestions** ~~Revision~~ ~~Rejected~~	
Suggestions/ Reasons/ Remarks • Revise the title as suggested by the committee • Change the hypothesis to research question • Revise the objectives	

Please note:

- Inform IEC/SRC immediately in case of any adverse events and serious adverse events.
- Inform IEC/SRC in case of any change of study procedure, site and investigator
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- Submit revised protocol if applicable



(Dr. Deekshit)
Member Secretary SRC

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28.07.2025

Ms. Manohari M Kulkarni

NU24BSCB03

At the research advisory committee meeting held on 26-07-2025 at NUCSER, your research proposal titled "**ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NSMCE2 - DEFICIENT BREAST CANCER CELLS"** was discussed and recommended. The scientific review committee hereby approves the proposal.

Deekshit
Member Secretary

Member Secretary
SRC, NUCSER

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CERTIFICATE

This is to certify that the M.Sc. Dissertation titled, "**ROLE OF COMPENSATORY SUMOYLATION MECHANISMS IN NSMCE2-DEFICIENT BREAST CANCER CELLS",** submitted by **MANOHARI M KULKARNI**, Reg.No: **NU24BSCB03** , Nitte University Centre for Science Education and Research, Paneer Campus, Deralakatte, has been subjected to TURNITIN software for Anti-Plagiarism and it is found to have similarity index of 1%.

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